

Screen-Space Perpetual Rendering of Skin

Jorge Jimenez, Veronica Sundstedt, and Diego Gutierrez

ACM Transactions on Applied Perception (TAP), Vol. 6, No. 4, Article 23

Date Published: September 2009

DOI: 10.1145/1609967.1609970

Diksha Chottani

Motivation

The why

- **Fascination** Always been fascinated with the technical aspect in the medical field.
 - **Focus** Also, the chosen research paper perfectly aligns with my dissertation work.
 - **Feasibility** The Algorithm allows interactive demo creation which is one of requirements.
-

Implementation

The how

- C++ and OpenGL for core pipeline and scene management.
 - GLSL fragment shaders for screen-space post processing.
 - Multi-pass separable Gaussian blur modulated by depth gradients.
 - Interactive UI for real-time adjustment of scattering and translucency parameters.
 - Dynamic toggling between standard diffuse shading and the screen-space subsurface scattering approximation.
-

Why Skin is hard to render?



Subsurface Scattering

Light penetrates skin, through layers of epidermis and dermis, and exits at a different point. Standard diffuse shading ignores this entirely, which makes it look like plastic

Real-Time Cost

Prior state of art required irradiance rendering + UV space blurring per character which cost scales linearly with character count on screen

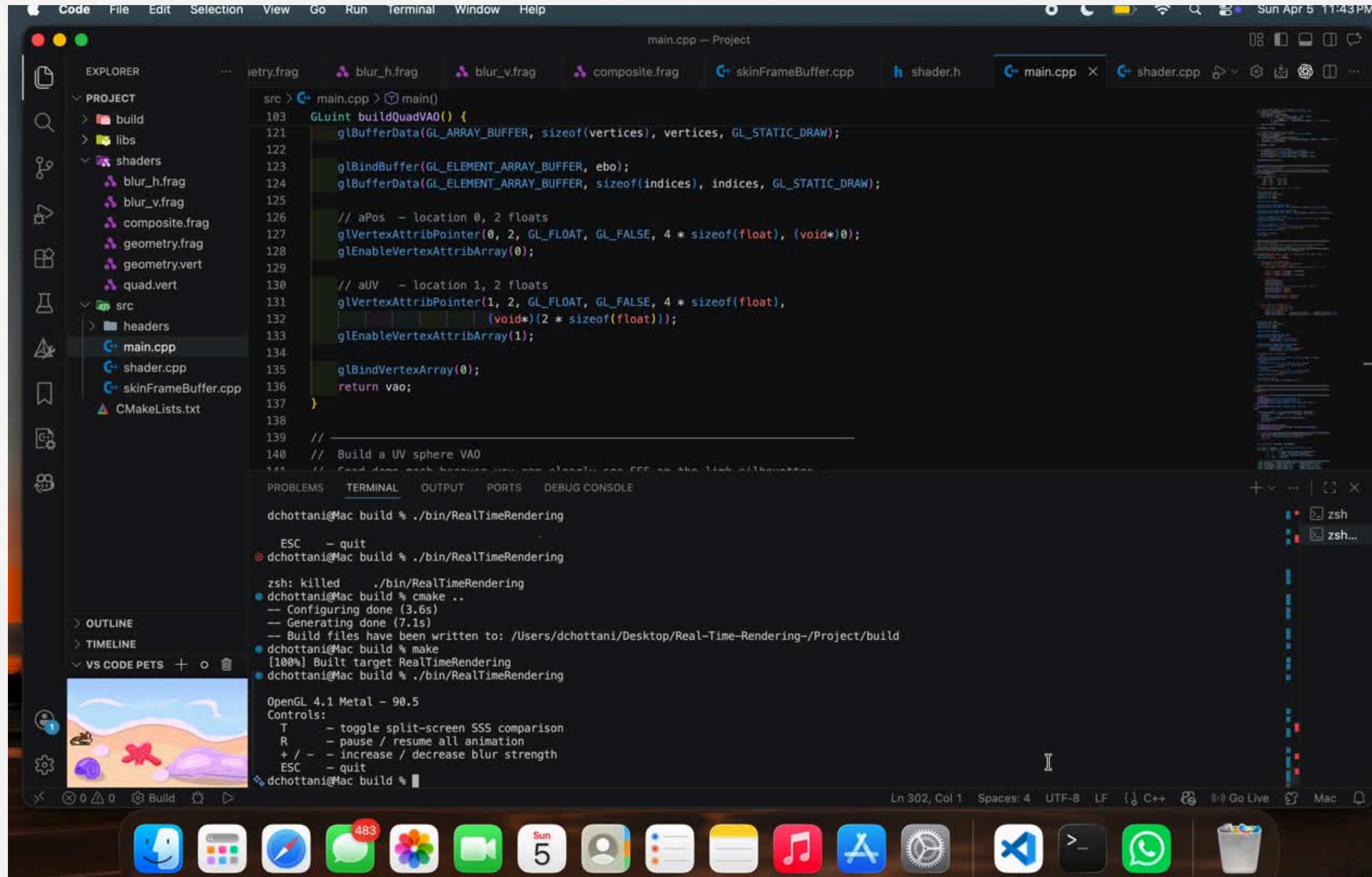
Texture-Space Limits

Texture-space methods requires a well parameterized UV unwrap and a dedicated texture per mesh, hard to integrate

What I learned building this?

- **Framebuffer Objects:** How to render to texture, chain multiple passes, and manage color + depth attachments in a multi-pass pipeline.
 - **Perceptual vs Physical:** The paper's psychological study showed screen-space approximation matches texture-space perceptually, even though it's lead physically correct.
 - **Diffuse/ Specular Splits:** Only diffuse light scatters; specular highlights stay sharp. Separating them in the G-buffer is foundational to the technique.
 - **Scalability:** Since SSS is screen-space post process, 1-10 characters costs the same, unlike texture-space methods that scale with character count.
-

Demonstration Video



Thank You!